

DIFFERENTIAL COHOMOLOGY SEMINAR 8

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The aim of this talk is to review twisted cohomology theory with the aim of later discussing twisted differential cohomology theories [BG16]. For these talks the main source is [ABG⁺14, ABG18].

1. TWISTED COHOMOLOGY

Let R be a ring spectrum, meaning a monoid object in the ∞ -category of spectra $\mathcal{S}p$. From this we get a presentable stable ∞ -category $\mathcal{M}od_R$ of left R -module spectra. Objects therein are morphisms of the form $R \wedge M \rightarrow M$ satisfying the usual associativity and unit conditions up to coherent homotopies.

Remark 1.1. Every stable category $\mathcal{C}$ is enriched over spectra. Given two objects C, D , we will denote by $\mathcal{H}om_{\mathcal{C}}(C, D)$ the hom-spectrum and by $\mathrm{Hom}_{\mathcal{C}}(C, D) = \Omega^{\infty} \mathcal{H}om_{\mathcal{C}}(C, D)$ the underlying hom-space.

Note we have an adjunction

$$\mathcal{S}p \begin{array}{c} \xrightarrow{R \wedge -} \\ \xleftarrow{\mathcal{H}om_{\mathcal{M}od_R}(R, -)} \end{array} \mathcal{M}od_R ,$$

where the right adjoint is equivalent to the forgetful functor.

Definition 1.2. Let R be a ring spectrum. An R -line is an R -module L such that $L \simeq R$.

Definition 1.3. Let $\mathcal{L}ine_R$ be the full sub- ∞ -groupoid of $\mathcal{M}od_R$ spanned by the R -lines.

By construction, $\mathcal{L}ine_R$ is equivalent to the category with a single object R and hom-space $GL_1 R$, the ∞ -group of R -linear automorphisms of R . Notice that $GL_1(R) \subseteq \mathrm{Hom}_{\mathcal{M}od_R}(R, R) \simeq \mathrm{Hom}_{\mathcal{S}p}(\mathbb{S}, R) = \Omega^{\infty} R$.

Lemma 1.4. $GL_1(R)$ fits into the pullback square

$$\begin{array}{ccc} GL_1(R) & \longrightarrow & \Omega^{\infty} R \\ \downarrow & & \downarrow \\ \pi_0(R)^{\times} & \longrightarrow & \pi_0(R) \end{array}$$

In particular, the inclusion $GL_1(R) \rightarrow \Omega^{\infty} R$ induces an isomorphism on n -homotopy groups, for all $n \geq 1$.

Proof. $\pi_0(R) \simeq \mathrm{Hom}_{\mathrm{hMod}_R}(R, R)$, where hMod_R is the homotopy category of R -modules, the right-vertical arrow corresponds to $\mathrm{Hom}_{\mathcal{M}od_R}(R, R) \rightarrow \mathrm{Hom}_{\mathrm{hMod}_R}(R, R)$ mapping a morphism to its homotopy class, and $\pi_0(R)^{\times} \subseteq \mathrm{Hom}_{\mathrm{hMod}_R}(R, R)$ is the set of isomorphisms. Finally, a morphism in a ∞ -category $\mathcal{C}$ is an equivalence if and only if its homotopy class is an isomorphism in $\mathrm{h}\mathcal{C}$. $\square$

Definition 1.5. Let X be a space. Denote by $\mathcal{M}od_R(X)$ the ∞ -category of R -module spectra parametrized over X , i.e. the functor category $\mathrm{Fun}(X^{\mathrm{op}}, \mathcal{M}od_R)$.

Definition 1.6. Let X be a space. Denote by $\mathcal{L}ine_R(X)$ the ∞ -groupoid of R -line spectra parametrized over X , i.e. the functor category $\mathrm{Fun}(X^{\mathrm{op}}, \mathcal{L}ine_R)$.

Since $\mathcal{L}ine_R \simeq BGL_1(R)$, functors $X^{\mathrm{op}} \rightarrow \mathcal{L}ine_R$ classify $GL_1(R)$ -principal bundles over X . In this sense, elements of $\mathcal{L}ine_R(X)$ are ∞ -local systems.

Example 1.7. Let $R_X: X^{\mathrm{op}} \rightarrow * \rightarrow \mathcal{L}ine_R$ be the constant functor with value R .

We now proceed to the Thom construction. We introduce it to full generality.

Definition 1.8 (Thom spectrum). The *Thom R -module spectrum* is the functor

$$R : \mathcal{S}/\mathrm{Mod}_R \rightarrow \mathrm{Mod}_R,$$

which sends $f : X^{\mathrm{op}} \rightarrow \mathrm{Mod}_R$ to the R -module spectrum R^f given by colimit of f .

Remark 1.9. The colimit of a functor $F : \mathcal{C} \rightarrow \mathcal{X}$ is the left Kan extension of F along the terminal functor $p : \mathcal{C} \rightarrow *$. If $\mathcal{X}$ is cocomplete, a functor $\pi : \mathcal{C} \rightarrow \mathcal{D}$ induces a pullback functor $\pi^* : \mathcal{F}\mathrm{un}(\mathcal{D}, \mathcal{X}) \rightarrow \mathcal{F}\mathrm{un}(\mathcal{C}, \mathcal{X})$, which is cocontinuous, therefore using left Kan extension along π we can construct a left adjoint $\pi_!$.

Let us have a look at some examples of Thom spectra.

Example 1.10 (Trivial twist). A functor $f : \mathcal{X} \rightarrow \mathcal{Y}$ induces a post-composition functor $f_* : \mathcal{S}/\mathcal{X} \rightarrow \mathcal{S}/\mathcal{Y}$ that is left adjoint to f^* sending $F : \mathcal{C} \rightarrow \mathcal{Y}$ to the canonical projection $\mathcal{X} \times_{\mathcal{Y}} \mathcal{C} \rightarrow \mathcal{X}$. If $\mathcal{X} = *$, we can identify $\mathcal{S}/\mathcal{X}$ with $\mathcal{S}$, and f with an object $E \in \mathcal{Y}$. If $\mathcal{Y}$ is cocomplete, there is a colimit functor $\pi_! : \mathcal{S}/\mathcal{Y} \rightarrow \mathcal{Y}$ and the composition

$$(1.11) \quad \mathcal{S} \xrightarrow{f_*} \mathcal{S}/\mathcal{Y} \xrightarrow{\pi_!} \mathcal{Y}$$

preserves colimits, therefore it is identified by the image of the one-point space, which is E . If $\mathcal{Y} = \mathrm{Mod}_R$, then f corresponds to a R -module E . The composition

$$(1.12) \quad \mathcal{S} \xrightarrow{\Sigma_+^\infty} \mathcal{S}\mathrm{p} \xrightarrow{E \wedge -} \mathrm{Mod}_R$$

also preserves colimits and maps the one-point space to E , therefore the Thom spectrum of E_X is $E \wedge \Sigma_+^\infty X$, for every space X .

Definition 1.13 (Twisted cohomology). Let R be a ring spectrum, X be a space, $p : X \rightarrow *$ the terminal functor, and $f : X^{\mathrm{op}} \rightarrow \mathcal{L}\mathrm{ine}_R$ a R -line bundle over X . The *f -twisted R -cohomology of X* is defined as the mapping spectrum

$$R_f(X) := \mathcal{H}\mathrm{om}_{\mathrm{Mod}_R}(Mf, R) \simeq \mathcal{H}\mathrm{om}_{\mathrm{Mod}_R(X)}(f, p^*R) \simeq \mathcal{H}\mathrm{om}_{\mathrm{Mod}_R(X)}(f, R_X).$$

Similarly, the *f -twisted R -homology of X* is defined as

$$R^f(X) := \mathcal{H}\mathrm{om}_{\mathrm{Mod}_R}(R, Mf) \simeq Mf.$$

The f -twisted R -cohomology groups of X are defined as the homotopy groups of $R_f(X)$, i.e.

$$R_f^n(X) := \pi_0(\mathcal{H}\mathrm{om}_{\mathrm{Mod}_R}(Mf, \Sigma^n R)) \cong \pi_{-n}(\mathcal{H}\mathrm{om}_{\mathrm{Mod}_R(X)}(f, R_X)).$$

Similarly, the f -twisted R -homology groups of X are defined as the homotopy groups of $R^f(X)$, i.e.

$$R_n^f(X) := \pi_0(\mathcal{H}\mathrm{om}_{\mathrm{Mod}_R}(\Sigma^n R, Mf)) \cong \pi_n(Mf).$$

Example 1.14 (Trivial twist). If $f : X \rightarrow \mathcal{L}\mathrm{ine}_R$ factors through $*$, then f factors as $X^{\mathrm{op}} \rightarrow * \rightarrow \mathcal{S}$, the constant factor with value $*$, and $R \wedge \Sigma_+^\infty(-) : \mathcal{S} \rightarrow \mathrm{Mod}_R$. The latter functor commutes with colimits, being a left adjoint, while the colimit of the latter is X itself, then $Mf \simeq R \wedge \Sigma_+^\infty X$. In particular, f -twisted R -cohomology and R -homology of X reduce to ordinary (untwisted) R -cohomology and R -homology of X .

2. EXAMPLES OF TWISTED COHOMOLOGY

We now proceed to analyze several examples of twisted cohomology theories. This requires some preliminary lemmas.

Lemma 2.1. Consider a space X and the ∞ -categorical Yoneda's embedding $y : X \rightarrow \mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{S})$. The colimit of y is the terminal pre-sheaf on X , i.e. the pre-sheaf with constant value the one-point space.

Proof. Let S be a pre-sheaf on X , consider then the slice category $X_{/S}$ of pairs (x, ϕ) , where x is an object of X and $\phi : y(x) \rightarrow S$. The density theorem for ∞ -categories states that S is equivalent to the colimit of $X_{/S} \rightarrow X \xrightarrow{y} \mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{S})$, the first map being the canonical projection. Take $S = *$, then $X_{/*} \rightarrow X$ is an equivalence, hence the claim. $\square$

Let G be a topological group and BG the ∞ -groupoid with a single object $*$ and hom-space G . The category $\mathcal{S}_G := \mathcal{F}\mathrm{un}(BG, \mathcal{S})$ is equivalent to the category of G -spaces.

Lemma 2.2. Consider $X = BG$, a G -space $f : X \rightarrow \mathbb{S}$ and its left Kan extension $f_! : \mathbb{S}_G \rightarrow \mathbb{S}$, then $f_! \simeq (- \times E)/G$, where $E = f(*)$.

Proof. Evaluate at $*$, then $f_!(y(*)) = E$, by definition, and $y(*) \simeq G$, as G -spaces, hence $(y(*) \times E)/G \simeq (G \times E)/G \simeq E$. Since $f_!$ and $(- \times E)/G$ agree on representables and are colimit-preserving, they are equivalent. $\square$

Example 2.3. Take the space $BO(n)$ and $f_n : BO(n) \rightarrow \mathbb{S}_*$ the n -sphere S^n with $O(n)$ -action coming from the one-point compactification of the regular action on $\mathbb{R}^n$. Let $\alpha_n = \Sigma^{\infty-n} f_n : BO(n) \rightarrow \mathbb{S}_p$, then $\alpha_n(*) = \Sigma^{\infty-n} f_n(*) = \Sigma^{\infty-n} S^n \simeq \mathbb{S}$, so α_n factors through $\mathcal{L}ine_{\mathbb{S}}$. Let $X = BO(n)^{op}$ and $p : X \rightarrow *$ the terminal functor, then

$$M\alpha_n = p_! \Sigma^{\infty-n} f_n \simeq \Sigma^{\infty-n} p_!(f_n)_! y \simeq \Sigma^{\infty-n} (f_n)_! \underbrace{p_!(y)}_{\simeq *}_{\simeq *} \simeq \Sigma^{\infty-n} (* \times S^n)/O(n)$$

Let $P = EO(n)$ be the universal $O(n)$ -bundle and M a $O(n)$ -space, then $* \times_{O(n)} M$ is modelled by the *strict* quotient $(P \times M)/O(n)$, then

$$S^n/O(n) = \text{cofib}(\mathbb{R}_0^n \subseteq \mathbb{R}^n)/O(n) \simeq \text{cofib}(\underbrace{* \times_{O(n)} \mathbb{R}_0^n}_{\simeq E_0^n} \subseteq \underbrace{* \times_{O(n)} \mathbb{R}^n}_{\simeq E^n}) = \text{Th}(E^n)$$

where $E^n = P \times_{O(n)} \mathbb{R}^n \rightarrow BO(n)$ is the universal n -dimensional vector bundle, hence $M\alpha_n \simeq \Sigma^{\infty-n} \text{Th}(E^n)$.

The functor $BO(n) \rightarrow \mathcal{L}ine_{\mathbb{S}}$ induces a ∞ -group homomorphism $j_n : O(n) \rightarrow GL_1(\mathbb{S})$, mapping ϕ to $\Sigma^{\infty-n} \text{Th}(\phi)$. Consider the suspension morphism $s_n = \mathbb{R} \oplus - : O(n) \rightarrow O(1+n)$, then

$$j_n(\mathbb{R} \oplus \phi) = \Sigma^{\infty-n-1} \text{Th}(\mathbb{R} \oplus \phi) \simeq \Sigma^{\infty-n-1} \underbrace{\text{Th}(\mathbb{R})}_{\simeq S^1} \wedge \text{Th}(\phi) \simeq \Sigma^{\infty-n} \text{Th}(\phi) = j_n$$

Recall that the colimit over the suspension morphisms s_n is the stable orthogonal group O .

Definition 2.4. Denote by j the induced group homomorphism $O \rightarrow GL_1(\mathbb{S})$, called the *J-homomorphism*.

Example 2.5. Let $X = O^{op}$ and take $Bj : BO \rightarrow \mathcal{L}ine_{\mathbb{S}}$, then Mj is denoted MO and called the *real bordism spectrum*.

Denote by M the extended Thom spectrum functor $\text{Grp}_{\infty}^{op}/\text{Mod}_R \rightarrow \text{Mod}_R$, this is a left adjoint to the functor sending a R -module to the corresponding functor $* \rightarrow \text{Mod}_R$. In particular, M preserves colimits and $Bj \simeq \text{colim}_n Bj_n$, therefore we have the following:

Theorem 2.6. $MO \simeq \text{colim}_n MO(n) = \text{colim}_n \Sigma^{\infty-n} \text{Th}(E^n)$.

Example 2.7. A group homomorphism $\xi : G \rightarrow O$ induces a functor $f : BG \rightarrow \mathcal{L}ine_{\mathbb{S}}$. The Thom spectrum Mf is denoted MG or $M\xi$, and called *G-bordism spectrum*. For $G = U, SO, Spin$, and *String*, we obtain the *complex, oriented, spin, and string bordism spectra*.

Remark 2.8. In [Equation \(2.7\)](#) we might take $G = \{*\}$, the one-point group, then $MG \simeq \mathbb{S}$, which, if it didn't have a name, might be called *framed bordism spectrum*, following the naming convention in [Equation \(2.7\)](#) and in line with the theorem that $\pi_*(\mathbb{S}) \simeq \Omega_*^{\text{fr}}$, the bordism ring of framed (trivialized tangent bundle) smooth manifolds.

Let R be a ring in sets, then R is a A_{∞} -ring spectrum (actually, E_{∞}), $\Omega^{\infty} R$ is equivalent to R with discrete topology ($\pi_0(\Omega^{\infty} R) \simeq R$, as sets, and every other homotopy group vanish). In particular, $GL_1(R)$ is simply $R^{\times}$ with discrete topology. Consider then the fiber sequence $SO \rightarrow O \rightarrow \mathbb{Z}^{\times} \simeq GL_1(\mathbb{Z})$.

Example 2.9. $X = BO^{op}$ and $\alpha = w_1 : BO \rightarrow \mathcal{L}ine_{\mathbb{Z}}$, the 1st Stiefel-Whitney class (delooping of the determinant $O \rightarrow GL_1(\mathbb{Z})$), then Mw_1 is a $\mathbb{Z}$ -module spectra. Let $i : SO \subseteq O$, then $w_1 i$ factors through the point, so $M(w_1 i) \simeq \mathbb{Z} \wedge \Sigma_+^{\infty} SO$.

Given $f : X^{op} \rightarrow \mathcal{L}ine_R$ and a sequence $F \xrightarrow{i} Y \xrightarrow{\pi} X$, there is an induced sequence of Thom R -module spectra $MF \rightarrow MY \rightarrow MX$. If πi factors through the point, $MF \simeq R \wedge \Sigma_+^{\infty} F$.

Lemma 2.10. Let R be a ring spectrum and X a connected monoidal ∞ -groupoid, then

$$\text{Hom}_{\text{Mon}(\mathbb{S}_p)}(\Sigma_+^{\infty} X, R) \simeq \text{Hom}_{\text{Mon}(\mathbb{S})}(X, GL_1(R))$$

Proof. Since X is connected, the space of homomorphisms $X \rightarrow GL_1(R)$ is equivalent to the space of homomorphisms $X \rightarrow \Omega^\infty R$, then use that $(\Sigma^\infty, \Omega^\infty)$ is a monoidal adjunction (The monoidal structure on spectra is such that Σ^∞ is strong monoidal). $\square$

Remark 2.11. Notice that we can weaken the result. Namely, if X is 1-connected (pointed and connected), then the space of functors (of ∞ -groupoids) $X \rightarrow GL_1(R)$ is equivalent to the space of functors $X \rightarrow \Omega^\infty R$ such that $* \rightarrow X \rightarrow \Omega^\infty R$ is an equivalence. This last space is equivalent, via the $(\Sigma_+^\infty, \Omega^\infty)$ adjunction, to the space of morphisms of spectra $\Sigma_+^\infty X \rightarrow R$, such that $\mathbb{S} \rightarrow \Sigma_+^\infty X \rightarrow R$ represents a unit in $\pi_0(R)$.

Remark 2.12. Notice that we can also strengthen the result. Namely, if X is a connected, commutative monoid object and R is a commutative ring spectrum, then $\Omega^\infty R$ and $GL_1(R)$ are also commutative monoid objects. Using the same argument, together with the fact that $(\Sigma^\infty, \Omega^\infty)$ is actually a *symmetric* monoidal adjunction, we conclude that

$$\mathrm{Hom}_{\mathcal{C}\mathrm{Mon}(\mathcal{S}\mathrm{p})}(\Sigma_+^\infty X, R) \simeq \mathrm{Hom}_{\mathcal{C}\mathrm{Mon}(\mathcal{S})}(X, GL_1(R))$$

Remark 2.13. Let $\mathcal{D}$ be a monoidal ∞ -category. Consider $\mathrm{Cat}_\infty/\mathcal{D}$, the ∞ -category of functors into $\mathcal{D}$, with monoidal structure given by

$$(F : \mathcal{A} \rightarrow \mathcal{D}, G : \mathcal{B} \rightarrow \mathcal{D}) \longmapsto (\mathcal{A} \times \mathcal{B} \xrightarrow{F \times G} \mathcal{D} \times \mathcal{D} \xrightarrow{\otimes} \mathcal{D})$$

The monoidal unit is the functor $* \rightarrow \mathcal{D}$ picking out the monoidal unit of $\mathcal{D}$. If $\mathcal{D}$ is symmetric monoidal, then so is $\mathrm{Cat}_\infty/\mathcal{D}$. A (commutative) monoid object in $\mathrm{Cat}_\infty/\mathcal{D}$ is given by a (symmetric) monoidal category $\mathcal{C}$ and a (symmetric) monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$.

In view of [Equation \(2.13\)](#), let R be commutative ring spectrum, then Mod_R is a symmetric monoidal ∞ -category and Line_R is a symmetric monoidal ∞ -groupoid. The category $\mathrm{Grpd}_\infty^{\mathrm{op}}/\mathrm{Line}_R$ is then symmetric monoidal and (commutative) monoid objects are given by (symmetric) monoidal ∞ -groupoids X^{op} a (symmetric) monoidal functors $X^{\mathrm{op}} \rightarrow \mathrm{Line}_R$. One can then check that M is a symmetric monoidal functor, so that (commutative) monoid objects are sent to (commutative) monoid objects in Mod_R , i.e. (commutative) R -algebras.

Example 2.14. Let Tmf be the commutative ring spectrum of topological modular forms (see [Equation \(2.15\)](#)) and $\sigma : MString \rightarrow \mathrm{tmf}$ the *String*-orientation of tmf . In the sequence

$$BString \longrightarrow BO \xrightarrow{Bj} \mathrm{Line}_{\mathbb{S}}$$

all functors are symmetric monoidal, so that $MString$ is a commutative $\mathbb{S}$ -algebra, i.e. a commutative ring. The *String*-orientation of tmf is also a commutative ring homomorphism. In the fiber sequence $K(\mathbb{Z}, 3) \rightarrow BString \rightarrow BSO$, the fiber map $i : K(\mathbb{Z}, 3) \rightarrow BString$ is also a symmetric monoidal, so the composition

$$\Sigma_+^\infty K(\mathbb{Z}, 3) \xrightarrow{Mi} MString \xrightarrow{\sigma} \mathrm{tmf}$$

is a commutative ring homomorphism. Using [Equation \(2.10\)](#), we conclude that the induced homomorphism $K(\mathbb{Z}, 3) \rightarrow \Omega^\infty \mathrm{tmf}$ (which is a homomorphism of commutative monoid objects, given [Equation \(2.12\)](#)) factors through $GL_1(\mathrm{tmf})$, and so it induces a (symmetric monoidal) functor $K(\mathbb{Z}, 4) \rightarrow \mathrm{Line}_{\mathrm{tmf}}$, i.e. *2-bundle gerbes twist* tmf .

Remark 2.15. The spectrum of topological modular forms comes in three main flavors, namely:

- (1) TMF, i.e. the global sections of the spectral structure sheaf $\mathcal{O}^{\mathrm{top}} : (\mathrm{Aff}/\mathcal{M}_{\mathrm{ell}})^{\mathrm{op}} \rightarrow \mathcal{C}\mathrm{Mon}(\mathcal{S}\mathrm{p})$ on the (étale site of the) moduli stack of elliptic curves.
- (2) Tmf , i.e. the global sections of the spectral structure sheaf $\bar{\mathcal{O}}^{\mathrm{top}} : (\mathrm{Aff}/\bar{\mathcal{M}}_{\mathrm{ell}})^{\mathrm{op}} \rightarrow \mathcal{C}\mathrm{Mon}(\mathcal{S}\mathrm{p})$ on the (étale site of the) *compactified* moduli stack of elliptic curves. The inclusion $\mathcal{M}_{\mathrm{ell}} \hookrightarrow \bar{\mathcal{M}}_{\mathrm{ell}}$ induces a commutative ring homomorphism $\mathrm{Tmf} \rightarrow \mathrm{TMF}$.
- (3) tmf , i.e. the connective cover of Tmf . By definition, there is a commutative ring homomorphism $\mathrm{tmf} \rightarrow \mathrm{Tmf}$.

In [\[AHR10\]](#), tmf is used to denote our TMF, in [\[Goe09\]](#), tmf is used to denote our Tmf , and [\[DFHH14\]](#) has the same notation as us. In [Equation \(2.14\)](#), we use tmf to mean the connective cover of Tmf .

Let us go down one step in the chromatic ladder.

Example 2.16. Recall the fiber sequences for $Spin$ and $Spin^c$:

$$\mathbb{Z}_2 \rightarrow Spin \rightarrow SO, \quad S^1 \rightarrow Spin^c \rightarrow SO$$

All the spaces involved are commutative groups. Applying the Thom spectrum functor to the delooped sequences, we get

$$\Sigma_+^\infty K(\mathbb{Z}_2, 1) \rightarrow MSpin \rightarrow MSO, \quad \Sigma_+^\infty K(\mathbb{Z}, 2) \rightarrow MSpin^c \rightarrow MSO$$

Let $\sigma : MSpin \rightarrow KO$ and $\sigma^c : MSpin^c \rightarrow KU$ be the Atiyah-Bott-Shapiro orientation of real and complex K -theory (see [ABS64]). Similar to Equation (2.14), we get homomorphisms

$$\Sigma_+^\infty K(\mathbb{Z}_2, 1) \longrightarrow MSpin \xrightarrow{\sigma} KO, \quad \Sigma_+^\infty K(\mathbb{Z}, 2) \longrightarrow MSpin^c \xrightarrow{\sigma^c} KU$$

Using Equation (2.10) again and delooping, we obtain functors $K(\mathbb{Z}_2, 2) \rightarrow \mathcal{L}ine_{KO}$ and $K(\mathbb{Z}, 3) \rightarrow \mathcal{L}ine_{KU}$, i.e. *real, resp. complex, bundle gerbes twist real, resp. complex, K -theory*.

3. TWISTS VIA PICARD GROUPOIDS AND GRADING

This section requires some further details. Up until now we defined everything via $\mathcal{L}ine_R$, however for many applications we need to work with $\mathcal{P}ic_R$ instead.

Definition 3.1. Given a monoidal ∞ -category $(\mathcal{C}, \otimes, 1)$, an object M is *invertible* if there is an object N such that $N \otimes M \simeq M \otimes N \simeq 1$. The *Picard ∞ -groupoid* of $\mathcal{C}$ is the sub- ∞ -groupoid generated by invertible modules.

Remark 3.2. A monoidal category $(\mathcal{C}, \otimes, 1)$ is *closed* if, for every object M , the *left tensoring with M* functor $M^\otimes : \mathcal{C} \rightarrow \mathcal{C}$ admits a right adjoint $F(M, -)$. If M is invertible, with inverse N , then $M^\otimes$ is an equivalence with $N^\otimes$ as inverse. In particular, we can promote $(M^\otimes, N^\otimes)$ to an adjoint equivalence. By uniqueness of adjoint functors $F(M, -) \simeq N^\otimes$, and so

$$N \simeq N \otimes 1 = N^\otimes(1) \simeq F(M, 1) =: DM$$

Definition 3.3. Given a closed monoidal category $(\mathcal{C}, \otimes, 1)$, the functor $D := F(-, 1) : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ will be called *duality*. Given an object M , the object DM is called the *dual of M* .

Definition 3.4. If R is a ring spectrum, Mod_R is closed monoidal. Denote by $\mathcal{P}ic_R$ the Picard ∞ -groupoid of Mod_R .

Remark 3.5. $\Sigma^n R$ is invertible, with inverse $\Sigma^{-n} R$. In particular, there is a map $\mathbb{Z} \times \mathcal{L}ine_R \rightarrow \mathcal{P}ic_R$. However, this map need not be neither injective (if R is n -periodic), nor surjective (see [HM17]).

As mentioned in, the Thom spectrum functor makes sense for every functor $f : X^{\text{op}} \rightarrow \text{Mod}_R$. However, all examples of twists encountered so far came from functors into $\mathcal{L}ine_R$. An example of twist that is not the result of a R -line bundle is the *degree shift*.

Definition 3.6. Denote by M the *Thom R -module spectrum* functor

$$\text{Grpd}_\infty^{\text{op}} / \text{Mod}_R \rightarrow \text{Mod}_R$$

sending a functor $f : X^{\text{op}} \rightarrow \text{Mod}_R$ to its colimit.

Example 3.7. Let $f : X^{\text{op}} \rightarrow \mathcal{L}ine_R$ be a twist. Denote by $\Sigma^n f$ the composition of f with the shift functor $\Sigma^n : \mathcal{L}ine_R \rightarrow \mathcal{P}ic_R$. Since Σ^n is an equivalence, it commutes with colimits, so

$$M\Sigma^n f \simeq \Sigma^n Mf$$

If $f = R_X$, then $M\Sigma^n f \simeq \Sigma^n R \wedge \Sigma_+^\infty X$, so $\Sigma^n f$ -twisted R -cohomology and R -homology correspond to normal R -cohomology and R -homology with a degree shift by n .

4. UMKEHR MAP

We now proceed to the construction of the umkehr map in twisted cohomology theories. Here we follow [ABG18]. Let R be a ring spectrum, denote by D_R the duality of Mod_R . Given an invertible R -twist $\alpha : X^{\text{op}} \rightarrow \text{Pic}_R$, denote by $D_R\alpha$ the post-composition of α^{op} with D_R .

Remark 4.1. Depending on which cohomological degrees we want, we sometimes use the following alternative definition of twisted cohomology:

$$R^\alpha(X) = \pi_0 \text{Hom}_{\text{Mod}_R}(M(D_R\alpha), R)$$

meaning we use the dual/inverse of α . To justify the use of $D_R\alpha$, consider the following: Let $\beta = D_R\alpha$, then

$$\text{Hom}_{\text{Mod}_R}(M\beta, R) \simeq \text{Hom}_{\text{Mod}_R(X)}(D_R\alpha, R_X) \simeq \text{Hom}_{\text{Mod}_R(X)}(R_X, \alpha \otimes_{R_X} R_X) \simeq \text{Hom}_{\text{Mod}_R(X)}(R_X, \alpha)$$

The second equivalence is a consequence of the fact that, if M be an invertible R -module, left tensoring with M is left adjoint to tensoring with D_RM , and $D_RD_RM \simeq M$, see Equation (3.2). If we think of α as a bundle of invertible R -modules over X , then $\text{Hom}_{\text{Mod}_R(X)}(R_X, \alpha)$ is the spectrum of global sections of α , which aligns with the idea that twisted cohomology is the homotopy groups of the global sections of a bundle of spectra.

If $R = \mathbb{S}$, we denote the duality D_R by simply D .

Definition 4.2. The *Spanier-Whitehead dual* of a space X is the dual spectrum of $\Sigma_+^\infty X$ with respect to the sphere spectrum.

Remark 4.3. Recall that $\mathcal{S}\mathfrak{p}$ is a closed symmetric monoidal category, with $\mathcal{H}\text{om}_{\mathcal{S}\mathfrak{p}}(E, -)$ being the right adjoint to the functor given by smash product with E . Let $(\mathcal{C}, \otimes, 1)$ a closed monoidal category, where $F(X, -)$ is the right adjoint to the functor given by left tensoring with X , then

$$\text{hom}_{\mathcal{C}}(-, F(1, E)) \simeq \text{Hom}_{\mathcal{C}}(1 \otimes -, E) \simeq \text{Hom}_{\mathcal{C}}(-, E)$$

Using Yoneda's lemma, we conclude that $E \simeq F(1, E)$, for all E .

Example 4.4. By Equation (4.3), the Spanier-Whitehead dual to $*$ is the sphere spectrum, since

$$D* = \mathcal{H}\text{om}(\Sigma_+^\infty *, \mathbb{S}) = \mathcal{H}\text{om}(\Sigma^\infty S^0, \mathbb{S}) = \mathcal{H}\text{om}(\mathbb{S}, \mathbb{S}) \simeq \mathbb{S}$$

Definition 4.5. Denote by ϕ the functor $\mathcal{S}^{\text{op}} \rightarrow \mathbb{S}/\mathcal{S}\mathfrak{p}$ from spaces to the category of spectra under $\mathbb{S}$, mapping X to the map of spectra

$$\phi(X) : \mathbb{S} \simeq D* \xrightarrow{Dp} DX$$

where p is the terminal map $X \rightarrow *$.

Definition 4.6. Denote by $\widehat{\text{Mfd}}_\infty$ the topological groupoid of closed smooth manifolds and diffeomorphisms. The set of diffeomorphisms are topologized by the weak C^∞ topology, see [Hir94]. Let Mfd_∞ be homotopy coherent nerve of $\widehat{\text{Mfd}}_\infty$.

Let B be a connected compact space and $\pi : X \rightarrow B$ a continuous function such that, for every $b \in B$, the fiber $\pi^{-1}(b) =: X_b$ is equipped with the structure of a closed smooth manifold, which varies continuously in B , in the sense of being classified by a functor

$$f : B \rightarrow \text{Mfd}_\infty$$

Given such a classifying map f , consider the following composition

$$B^{\text{op}} \longrightarrow \text{Mfd}_\infty^{\text{op}} \longrightarrow \mathcal{S}^{\text{op}} \xrightarrow{\phi} \mathbb{S}/\mathcal{S}\mathfrak{p}$$

where the middle functor is the one forgetting the smooth structure. The above composition is then an object of

$$\mathcal{F}\text{un}(B^{\text{op}}, \mathbb{S}/\mathcal{S}\mathfrak{p}) \simeq \mathbb{S}_B / \mathcal{F}\text{un}(B^{\text{op}}, \mathcal{S}\mathfrak{p})$$

i.e. it is a natural transformation

$$\phi_{X/B} : \mathbb{S}_B \rightarrow D_B(f)$$

where D_B denotes the Spanier-Whithead duality applied to (the opposite of) $f : B \rightarrow \mathcal{S}$ point-wise. Let $p : B \rightarrow *$ be the terminal functor and $p_! : \mathcal{F}un(B^{\text{op}}, \mathcal{S}p) \rightarrow \mathcal{S}p$ the left Kan extension along p , i.e. the colimit functor. Applying $p_!$ to $\phi_{X/B}$, we obtain a morphism of spectra

$$(4.7) \quad \Sigma_+^\infty B \simeq p_! p^* \mathcal{S} = p_! \mathcal{S}_B \xrightarrow{p_!(\phi_{X/B})} p_! D_B(f)$$

Let $T_{X/B} \rightarrow X$ be the vector bundle of fiber-wise tangent vectors, classified by $X^{\text{op}} \rightarrow BO(n)$, where n is the manifold dimension of the fibers¹. Denote by α the composition

$$(4.8) \quad X^{\text{op}} \xrightarrow{T_{X/B}} BO(n) \longrightarrow \mathcal{S}_* \xrightarrow{\Sigma^\infty} \mathcal{S}p$$

the middle arrow being the n -sphere S^n with $O(n)$ -action coming from the one-point compactification of the regular action on $\mathbb{R}^n$. The above diagram takes values in the Picard ∞ -groupoid of the sphere spectrum.

Theorem 4.9. $p_! D_B(f) \simeq M(-\alpha) =: X^{-T_{X/B}}$.

Definition 4.10. The morphism

$$\mathcal{PT}(f) : \Sigma_+^\infty B \longrightarrow X^{-T_{X/B}}$$

is called *Pontryagin-Thom transfer map*.

Consider now a general situation. Let R be a ring spectrum and $\alpha : X^{\text{op}} \rightarrow \mathcal{L}ine_R$ a R -line bundle.

Definition 4.11. An *orientation* of $M\alpha$ is lift

$$\begin{array}{ccc} & & * \\ & \nearrow & \downarrow \\ X^{\text{op}} & \xrightarrow{\alpha} & \mathcal{L}ine_R \end{array}$$

Explicitly, it is an equivalence of R_X -modules $\alpha \rightarrow R_X$.

Applying $p_!$, we see that an equivalence $t : \alpha \rightarrow R_X$ induces an equivalence

$$M\alpha \xrightarrow{p_!(t)} R \wedge \Sigma_+^\infty X$$

In our case, α in Equation (4.8) is not valued in $\mathcal{L}ine_{\mathcal{S}}$, but $\Sigma^{-n}\alpha$ is. An orientation of $\Sigma^{-n}\alpha$ induces an equivalence

$$(4.12) \quad \Sigma^{-n} M\alpha \xrightarrow{\simeq} R \wedge \Sigma_+^\infty X$$

Definition 4.13. The isomorphism in cohomology induced by Equation (4.12) is called *Thom isomorphism*.

Finally, we have the following definition:

Definition 4.14. Assuming $\Sigma^{-n}\alpha : X^{\text{op}} \rightarrow \mathcal{L}ine_R$ is orientable, the *Umkehr map* is the map:

$$R^*(\Sigma_+^\infty X) \xrightarrow{\text{Thom iso.}} R^{*-n}(X^{-T_{X/B}}) \xrightarrow{\mathcal{PT}(f)^{*-n}} R^{*-n}(\Sigma_+^\infty B)$$

Remark 4.15. The name *Umkehr* comes from the map going in the opposite direction to $R^*(\Sigma_+^\infty X) \rightarrow R^*(\Sigma_+^\infty B)$, given by pulling back along π .

We now introduce the *twisted Umkehr map*. Given a twist $\beta : B^{\text{op}} \rightarrow \mathcal{P}ic_R$, we can smash to get a map

$$\beta \simeq \mathcal{S}_B \wedge_B \beta \longrightarrow D_B(f) \wedge_B \beta$$

Applying the functor $p_!$ once again, we obtain a morphism

$$M\beta \longrightarrow p_!(D_B(f) \wedge_B \beta) \simeq X^{-T_f + \beta\pi}$$

where $\beta\pi : X^{\text{op}} \rightarrow \mathcal{P}ic_R$ is the composition of π and the twist β .

Definition 4.16. The *twisted Pontryagin-Thom transfer map* is the morphism:

$$\mathcal{PT}(f, \beta) : M\beta \longrightarrow X^{-T_f + \beta\pi}$$

¹ B is connected, so the fibers have constant dimension.

The *twisted Umkehr map* is the map of twisted cohomology groups induced by $\mathcal{PT}(f, \beta)$.

Example 4.17. Assume that $X^{-T_{X/B}+\beta\pi}$ is orientable, i.e. the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{-T_f} & \mathrm{Pic}_{\mathbb{S}} \\ \downarrow & & \downarrow \\ B & \xrightarrow{\alpha} & \mathrm{Pic}_R \end{array}$$

then $X^{-T_{X/B}+\beta\pi} \simeq R \wedge \Sigma_+^\infty X$ and the twisted Umkehr map becomes:

$$R^*(\Sigma_+^\infty X) \longrightarrow R^{*- \beta}(B)$$

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